Problem 4: The necessity of Non-linear activations

- 3 layer neural network without activation function
$\begin{cases} \text{input} \rightarrow \text{vector } x \\ \text{first hidden layer} \rightarrow W^{[1]} \; b^{[1]} \\ \text{output} \rightarrow W^{[2]} \; b^{[2]} \end{cases}$

a) $z^{[1]} = x W^{[1]} + b^{[1]}$

b) $\hat{y} = z^{[1]} W^{[2]} + b^{[2]}$

c) from part a) $\hat{y} = z^{[1]} w^{[2]} + b^{[2]} \xrightarrow{z^{[1]} = x W^{[1]} + b^{[1]}}$

$\hat{y} = (x W^{[1]} + b^{[1]}) w^{[2]} + b^{[2]}$

d) Show above expression can be written as $\hat{y} = W' x + b'$

$\rightarrow \hat{y} = (x W^{[1]} + b^{[1]}) w^{[2]} + b^{[2]}$

$\hat{y} = \underbrace{x W^{[1]} w^{[2]}}_{W' = w^{[1]} w^{[2]} \text{ multiplier}} + \underbrace{b^{[1]} w^{[2]} + b^{[2]}}_{b' = b^{[1]} w^{[2]} + b^{[2]} \text{ constant}}$

$\hat{y} = W' x + b'$

e) Non linear activation functions are essential to give NN their representational power as without these activation functions, even if we have multiple NN layers like this example, they will be turned to one linear layer and will be equivalent to single linear transformation $\hat{y} = W' x + b$.
By adding non-linear activation functions, NN model can learn complex boundries.
(composition of linear maps is a linear map. A 3-layer network has same representation power of a single linear layer.)